

Safety Case Addendum AURIX™ TC39x BD-Step Microcontroller



Distribution under NDA only

About this document

About this document

Scope and purpose

This document provides the addendum to the previously released TC39x BD-Step Microcontroller Safety Case Addendum V1.0.

The product AURIX™ TC39x BD-Step Microcontroller described in this Safety Case Addendum has been developed and is released as ISO 26262:2018-compliant Safety Element out of Context (SEooC) for safety requirements up to ASIL D.

If the product is intended to be used in a functional safety application, you are obliged to ensure conformity with the relevant requirements of ISO 26262:2018.

In particular you are obliged to obtain the available product safety documents and to respect the information and information updates provided by Infineon Technologies AG.

In the event that you merely resell the product as a distributor, you are obliged to ensure that your customers obtain the available product safety documents and that your customers respect the information and information updates provided by Infineon.

Information updates are provided via Product Change Notification (PCN), Info Note or Early Problem Notification (EPN) or any other written communication to customers or distributors that have ordered the product within the last 5 years. The complete updated safety document is available via the Infineon sales office.

Intended audience

This Safety Case Addendum is written for experienced project managers, system integrators and functional safety managers involved in the development of a safety-related system who consider integrating the AURIX™ TC39x BD-Step Microcontroller into their application.

Scope

This document is the Safety Case Addendum for the AURIX™ TC39x BD-Step Microcontroller.

It serves as a summary of the work product Safety Case compiled during the product development phase at Infineon and of the individual Audit and Assessment report(s).

About this document

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Executive Summary

Executive Summary

This document describes the changes to the last release of TC39x BD-Step Microcontroller Safety Case Addendum V1.0. Since last release all received changes were analyzed for applicability to the TC39x BD-Step Microcontroller Safety Case Addendum V1.1. Changes were assessed and necessary modification were carried out as per the standard configuration management practice.

Necessary internal audits and assessments confirmed that the original compliance as stated in previously released TC39x BD-Step Microcontroller Safety Case Report V1.0 and TC39x BD-Step Microcontroller Safety Case Addendum/Addenda starting from Addendum version V1.0 is still valid. For details regarding all applicable Addendum versions please see revision history.

The purpose for the release of this Safety Case Addendum is to extend the initial claim for a compliant development according to ISO 26262:2011 to ISO 26262:2018.

While the hardware and software of the developed product Aurix TC39x BD-Step Microcontroller remains unchanged, changes between the first and second edition of ISO 26262 have been analyzed and, where found applicable, activities have been performed to provide necessary additional evidence. For analyzed changes where the available evidence has been found sufficient, an argument is provided.

In summary, this Safety Case Addendum does not address a product development but compliance to a newer version of the ISO 26262 only.

With release of this addendum, application of TC39x BD-Step Microcontroller Safety Case Report V1.0 and previously released TC39x BD-Step Microcontroller Safety Case Addendum/Addenda starting from Addendum version V1.0 shall be considered together with this addendum unless otherwise specified. For details regarding all applicable Addendum versions please see revision history.

Required additional reading

The product variants are covered by the SPRN. Technical descriptions such as safety mechanisms, diagnostic coverages and failure rates are documented in the Safety Manual and the FMEDA.

A list of the relevant documents can be found in **Chapter 7**.

Changes

Changes

Changes from Safety Case Addendum V1.0 to Addendum V1.1

The user of this document has to read this document version in conjunction with all previously released document versions including the released Safety Case Report document.

Meaning of bracketed terms below:

- 1.) (Modified): A similar entry was already present.
 - a. The corresponding information of the previous version of the Safety Case Addendum is updated, the modification is with respect to the chapter in the initial Safety Case Report and has to be read considering the previous version of the Safety Case Addendum.
 - b. Version information shows the last version of the Safety Case Addendum where a change has been applied.
- 2.) (New): This entry is new.
 - a. The corresponding information is related to the released version of the Safety Case Report document.

Example: (Modified; V1.0) Last change of the corresponding entry in the table below is documented in Safety Case Addendum V1.0 and additional change(s) documented within this document (Safety Case Addendum with version number > V1.0).

Table A: Overview of changes (Safety Case Addendum V1.0 to Addendum V1.1)

Safety Case Report Structure	Changes
Chapter 1	No changes
Chapter 2 (Modified; V1.0)	Changes in Chapter 2.1 – See details below.
Chapter 3	No changes
Chapter 4	No changes
Chapter 5 (Modified; V1.0)	Changes in Chapter 5.4 – See details below
Chapter 6	No changes
Chapter 7 (Modified; V1.0)	Changes in Chapter 7 – See details below.
Chapter 8 (Modified; V1.0)	Changed in Chapter 8 – See details below.
Chapter 9	No changes
Chapter 10	No changes

Details of changes according to the table above:

Chapter 2.1 (Modified compared to Safety Case Addendum V1.0)

Changes

This Safety Case Addendum applies to the *TC39x BD-Step Microcontroller* product variants given in below table.

Table 1 TC39x BD-Step Microcontroller product variants and derivatives

Name	Description and features	Package	Marking
See Safety Package Release Note	See Safety Package Release Note	See Safety Package Release Note	See Safety Package Release Note

This Safety Case Addendum extends the claims from the TC39x BD-Step Microcontroller Safety Case Addendum V1.0 to the variants listed in the table above and extends the initial claim for a development according to ISO 26262:2011-compliant to ISO 26262:2018-compliant.

Any claim on initially listed product variants in previous version remains valid unless otherwise explicitly noted.

Chapter 5.4 (Modified compared to Safety Case Addendum V1.0)

Table 5 Performed Audit and Assessments

Type	Scope	Start Date	End Date
Production Release Assessment (ISO26262)	<i>TC39x BD-Step Microcontroller & SBST</i>	2021-05-03	2021-05-27

All findings observed during the development phase have been adequately closed and therefore the final assessment result was rated as "PASS".

The following ISO 26262 confirmation reviews have been conducted:

Table 1 Performed confirmation reviews (2018)

Type	Infineon work product name	Scope
Confirmation review of the safety plan	Safety Plan - Confirmation Review Report	AURIX™ AURIX2G Family Microcontroller
Confirmation review of the technical safety concept	Safety Concept - Confirmation Review Report	AURIX™ AURIX2G Family Microcontroller
Confirmation review of the integration and test strategy	Product Verification Plan - Confirmation Review Report	AURIX™ AURIX2G Family Microcontroller
Confirmation review of the safety analyses and the	Safety Analysis - Confirmation Review Report	TC39x BD-Step Microcontroller

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Type	Infineon work product name	Scope
dependent failure analyses		
Confirmation review of the safety case	Safety Case - Confirmation Review Report	TC39x BD-Step Microcontroller

All findings observed during the confirmation reviews have been adequately closed. The other confirmation reviews are not applicable for the product.

Chapter 7 (Modified compared to Safety Case Addendum V1.0)

Table 8 "Reference List of Work Products and Internal Documents" in the TC39x BD-Step Microcontroller Safety Case Report V1.0 remains unchanged even though the Work Products and Internal Documents are newly baselined.

Table 9 "Reference List of Customer Documentation and Internal Documents" with customer relevant documentation is applicable for all product variants covered in initial Safety Case Report V1.0, Safety Case Addendum/Addenda starting from Addendum version V1.0 as well as in this Safety Case Addendum. For details regarding all applicable Addendum versions please see revision history.

Changes

Table 9 Reference List of Customer Documentation and Internal Documents

Document	Revision Number & Release Date
AURIX™ TC3xx IFX User Manual	See Safety Package Release Note
AURIX™ TC39x Appendix to User's Manual	See Safety Package Release Note
AURIX™ TC3xx ED Target Specification ¹	See Safety Package Release Note
TC39xB_DS	See Safety Package Release Note
TC39x_Data_Sheet_Addendum	See Safety Package Release Note
AURIX TC3xx Safety Manual	See Safety Package Release Note
TC39xBD_Errata_Sheet	See Safety Package Release Note
AURIX_TC39x_FMEDA	See Safety Package Release Note
CPU SBST User Manual	See Safety Package Release Note
SPU SBST User Manual	See Safety Package Release Note
A2G_AURIX Safety Analysis Summary Report	See Safety Package Release Note
TC39xBD_SafetyCaseAddendum	See Safety Package Release Note
TC39xBD_SafetyPackageReleaseNote	See Safety Package Release Note
TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	See Safety Package Release Note
TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	See Safety Package Release Note

¹ Distribution under NDA, only relevant for tool development not for application development

Changes

Chapter 8 (Modified compared to Safety Case Addendum V1.0)

The following table superseeds any previously communicated information and shall be carefully read and assessed by all relevant stakeholders. IFX assumes that the information contained in this table is forwarded to all relevant parties within the organization of the recipient of this document.

Table 10 Open Problems List

No.	Open Problem Description	Argument for Acceptance
1	EVRC Frequency and Phase Synchronization to CCU6/GTM input. As documented in section "EVRC Frequency and Phase Synchronization to CCU6/GTM Input" in TC3xx UM V1.x.0: Loss of Synchronization Lock event is triggered if switching frequency range is violated. Loss of Synchronization Lock event is indicated via status bits and interrupt. If this functionality is used in a safety related application, the faults leading to loss of synchronization could be detected by reading the status bit or reacting on the interrupt. This ESM is not part of the Safety Manual.	Current FMEDA provides FIT rate for faults of this functionality if configured to be used in safety related application. This is such small that FIT targets are not violated.
2	Customer documentation (FMEDA) does not provide evidence for usage of GPT12 in safety related applications in combination with GTM. However, Safety Manual still shows ESM which refer to GPT12. It is still possible for the integrator to use GPT12 upon own assessment.	Current FMEDA provides FIT rate for faults of GPT12 functionality if used in safety related application. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
3	Customer documentation (FMEDA) does not provide evidence for usage of PORTS as single channel in safety related applications. It is still possible for the integrator to use PORTS as single channel upon own assessment.	Current FMEDA provides FIT rate for faults of PORTS functionality if used as single channel in safety related applications. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
4	Safety Case Report 1.0 is baselined to a requirement database which has minor number of known incorrect relations of safety requirements (< 1%).	Defendable, all incorrect relations analyzed and assessed, no safety related issues found.
5	Risk of input degradation under humidity/temperature cycling stress.	Comprehensive safety impact analysis provides evidence, that occurrence of observed fail in the field will not compromise the product safety related aspects.

Revision history

Document version	Date of release	Description of changes
V1.0	2020-03	Initial Release
V1.1	2021-05	Refer to Table A

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